Introduction to Chemical Engineering Thermodynamics

Note: please check Brightspace regularly for updates to this syllabus.

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Classes: Lectures MWF 9:30-10:20am FRNY G140

Recitation:

Monday	4:30 - 5:20	SC G002
Monday	4:30 - 5:20	KNOY B041
Tuesday	3:30 - 4:20	HAMP 1266
Tuesday	4:30 - 5:20	HAMP 1266
Wednesday	3:30 - 4:20	HAMP 1266
Thursday	3:30 - 4:20	HAMP 1266

Teaching Assistants

-Miranda T. Rubio (rubio18@purdue.edu)
Office hours: Friday 2:30 pm-4:00 pm (FRNY B142A)
-Charlton Sullivan (sulli479@purdue.edu)
Office hours: Tuesdays 12:30-2:00 pm (FRNY B142A)
-Nafisa Arfa (narfa@purdue.edu)
Office hours: Friday 4 pm-5:30 pm (FRNY B142A)
-Chao-Hsiang Chang (chan1104@purdue.edu)
Office hours: Thursday 10:30 am-12:00 pm (FRNY B142A)

Course Description: Basic principles and concepts of thermodynamics applied to chemical engineering problems; use of basic thermodynamic functions of internal energy, enthalpy, entropy, and free energy to solutions, phase equilibria, chemical equilibria; thermodynamic processes; equations of state; and relation of macroscopic to molecular properties.

Textbook: CHE21100: Introductory Chemical Engineering Thermodynamics, Milo D. Koretsky . zyBook (ISBN: 979-8-203-14998-5), PURDUECHE21100SPRING2026. *This version allows students to print the text, students may purchase a printed version if the wish.*

See the instructions below to get access to the book on the zybook website: The subscription is $64.

1. Sign in or create an account at learn.zybooks.com
2. Enter zyBook code: PURDUECHE21100SPRING2026
3. Subscribe

Course Objectives: Develop a fundamental understanding of the key principles of macroscopic thermodynamics and apply this understanding to solve problems of practical importance in chemical engineering and allied fields of engineering and science.

Course Outcomes:

1. Understand the interconnection between work and heat.
2. Use the First Law of thermodynamics and energy balances in the analysis of closed and open systems.
3. Use the Second Law of thermodynamics and the application of the concept of entropy in the analysis of reversible and irreversible processes.
4. Derive and apply thermodynamic relations and relationships between thermodynamic potentials.
5. Analyze thermodynamic power and refrigeration cycles.
6. Predict, understand and apply the properties and phase equilibrium behavior of ideal and non-ideal fluids.
7. Understand the concepts of fugacity and chemical potential.
8. Apply thermodynamic principles to chemical reactions and equilibrium.
9. Practice clear, effective and concise written and oral communication of problem solutions.
10. Use computational tools in the solution of thermodynamics problems.

Course Topics: First Law of Thermodynamics; Second Law of Thermodynamics; estimation of physical properties; phase equilibria; chemical equilibria.

The course Brightspace website will serve as the information source for the class schedule and announcements. Lectures notes, supplementary reading materials, homework assignment, and grades will be posted online. The Brightspace site can be accessed from https://purdue.brightspace.com.

Course Structure: The *typical* weekly schedule will be:

Lectures: Monday, Wednesday, and Friday. The lectures will be delivered in-person in FRNY G140.

Recitation sessions: Recitation sessions will be available each week to review the basic concepts covered in lectures and help with problem solving. The recitation session will be supervised by course TA's. Recitations will be in person for all students. The recitation locations and times are given above.

Office hours will be conducted in person. If students need to quarantine, then a Webex version could also be implemented.

Assessment:
There are **three** midterm exams. It is anticipated that the midterms will be in-person. The exams will take place on:
(Exam 1: Monday, February 16, 2026; 9:30-10:20 AM FRNY G140)
(Exam 2: Wednesday, March 11, 2026; 9:30-10:20 AM FRNY G140)
(Exam 3: Friday, April 24, 2026; 9:30-10:20 AM; FRNY G140)

There are **weekly** homework assignments. The final grade will be calculated as:
Midterm 1: 29%, Midterm 2: 29%, Midterm 3: 29%, Homework 13%.
The course will be graded as follows. Some cut offs **may** be adjusted depending on the difficulty of the assessment items.

Grade	GPA Value	Recommended Range
A+	4.0	≥90%
A	4.0	≥87%
A-	3.7	≥85%
B+	3.3	≥80%
B	3.0	≥77%
B-	2.7	≥75%
C+	2.3	≥72%
C	2.0	≥70%
C-	1.7	≥67%
D+	1.3	≥65%
D	1.0	≥62%
D-	0.7 – becomes lowest passing grade	≥60%
F	0.0	< 60.0

Lecture Notes and Textbook: Additional lecture notes, other than the textbook, will be posted on the Brightspace site if necessary. For each section of the interactive textbook, we ask that you complete the online assessments before deadlines that we will announce in class. Students who complete greater than 80% of the assessments for the semester will have the option of dropping their lowest two homework grades at the end of the semester. This is the *only* means by which homework grades may be dropped.

Classroom Activities: We will periodically may solicit class feedback via iClicker Cloud. The primary purpose of the iClicker Cloud questions is to gauge student understanding of concepts in class.

Exams: The three midterms will primarily cover materials discussed in the preceding weeks.

Make-up Exams: Students are strongly urged to make attendance at exams a very high personal priority and make appropriate arrangements to be present at all exams. If an exam is missed students may choose to receive a zero for that assessment. Alternatively, **a single comprehensive make-up exam** (available only for students who miss a test for legitimate reasons) will be administered during the final exam period. This will be the **only** make-up exam available.

Homework: Homework for all students will be distributed on Brightspace and collected via Gradescope. Homework assignment will be handed out on Friday and due before class the next Friday. **No late homework will be accepted after 9:45 am on Friday**, without previous

permission of the instructor. You can **only** submit your own homework.

Attendance: Please plan to attend lectures. Numerous studies have shown that participation in class leads to better learning and course outcomes. Although attendance is not a formal part of your grade, it may be used to adjust grades if you are close to a cutoff at the end of the semester.

Re-grading policies: Any requests for re-grading homework or exams should be directed to the course instructors **within 1 week** after the requested materials are released to the student. No regrading requests will be considered beyond the one-week timeline. For any re-grading requests, we will re-grade the entire piece of work and you may end up gaining or losing points based on your overall performance.

On-line Course Evaluation: It is important for department and instructors to receive thorough feedback on all courses taught, so it is your responsibility to provide such feedback. Participation in the on-line course evaluation is mandatory and will be worth extra points if the participation rate is greater than 95%.

E-mail correspondence: expect a minimum 24-hour turn-around time for any e-mail correspondence. No homework related questions will be addressed by e-mail. Please make efficient use of all available office hours.

Ethical Conduct: Students in CHE 211 are embarking on a program of study to become chemical engineers and are expected to uphold the code of ethics, including

Honesty on homework. Your homework answers must reflect your own independent work and thinking. Discussion about homework problems is allowed and encouraged. However, do NOT copy homework and do NOT assist others by making your homework answers available. Any student detected copying or assisting in copying will receive zero credit on the assessment piece.

Honesty on exams. You are expected to work by yourself on problem sets during exams. No discussion is permitted on the relevant materials. You can only refer to allowed material as per specified in respective exams. All cell phones and mobile devices must be TURNED OFF during the test. Violations will result in, at least, a failing grade on the assessment piece. Serious cases will result in a failing grade for the course and documentation will be sent to the Dean of Students Office.

No disruption of class. Please turn off cell phones before lecture and recitation sessions. No phone calls and text messaging are allowed in class.

Syllabus – Tentative Schedule

Introduction and basic definitions (Chapter 1)	1 week (1/12-1/16)
Heat, work and 1st law of thermodynamics (Chapter 2)	3 weeks (1/21-2/11)
Entropy and 2nd law of thermodynamics (Chapter 3)	2 weeks (2/13-3/2)
Equation of states (Chapter 4)	1 week (3/4-3/13)
Fluid properties (Chapter 5)	1+ week (3/23-3/30)
Phase equilibria (Chapter 6 & 7, part of Chapter8)	3 weeks (4/1-4/22)
Chemical reaction equilibria (Chapter 9)	1+ weeks (4/24-5/1)

Note: Holidays and Class Recess Dates when there are no classes –
January 19, 2026 – Martin L. King Holiday
March 16-21, 2026 – Spring Break

This syllabus is subject to revision as necessary. Every effort will be made to maintain consistent procedures throughout the semester, but in extreme circumstances, some modifications may need to be made.